

Darden Consulting Club Case Book

2014-2015 Edition

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Compiled and edited by
Joy Arcangeli '15

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The Consulting Case Interview

Structure of the Case Interview

The case interview typically lasts between 20-40 minutes and consists of:

- Case overview/ prompt
- Clarifying questions
- Structure/ framework building & explanation
- Problem solving & analysis
- Recommendation

More information on each of these sections follows.

In general, the bulk of your time should be spent on problem solving & analysis, with 2-3 minutes to build your framework and a <2 minute recommendation at the end.

The interview should be highly conversational and interactive. Depending on the firm, round, and seniority of your interviewer, some interviews will be more structured than others. Please see the subsequent section for more firm-specific information.

Structure of the Case Interview

Case Overview/ Prompt

- The case interview will start with the interviewer providing the overview of the client and the problem. Take diligent notes, as the overview contains crucial information you will need later in the case.
- Following the prompt, you should give the interviewer your initial synthesis of the case. Confirm the objective and the key facts. Don't simply repeat back word-for-word what the interviewer said – this is the first chance to prove that you can synthesize information quickly
- Occasionally, the interviewer will not frame a specific question – one way to handle this is to test the client's goals with clarifying questions and propose an approach before structuring the problem

Clarifying Questions

- After the initial synthesis, you have the opportunity to ask some clarifying questions. This can be used to clarify a specific term used in the case prompt, to identify a concrete benchmark or goal for the problem(e.g. in a profitability case, can ask whether there are targets or time frames that the client has in mind) or to understand more about the client, their customer base, their competition, and their industry generally. Do not ask specific questions about the client at this stage, and do not ask more than 3-5 clarifying questions.

Structure of the Case Interview

Structure/ Framework Building & Explanation

- Following the prompt and clarifying questions, you should ask for some time to sketch out your framework
- Generally it is a good idea to write the objective at the top of your framework, and then structure it like an issue “tree”, with 2-4 buckets and a few supporting points underneath each
- Do not take more than 2 minutes to draw out your framework; then present to your interviewer
- It can be useful to read books like *Case in Point* that provide a bank of suggested frameworks, but **it is not recommended to try to memorize frameworks** as you run the risk of overlooking what is unique about each case situation and appearing robotic and overprepared

Problem Solving & Analysis

- After talking through the framework with your interviewer, propose the area of analysis that you’d like to start with. Do not wait for the interviewer to prompt you with “Where would you like to start?”.
- At this point your interviewer may suggest a different path or present you more information, especially data. If the interviewer does neither of those things, you should continue down the path you proposed and start identifying potential issues and areas of analysis, and then ask if the interviewer has any related data
- In general, the analysis will start off with some calculations or chart-reading, and then will move into a more qualitative/strategic direction where you’re asked to come up with some ideas. Keep in mind that this whole interview should be conversational in nature and listen carefully for hints.

Structure of the Case Interview

Recommendation

- Once you have worked through analysis and uncovered the main issues of the case, your interviewer will ask for a recommendation.
- You can ask for a minute to gather your thoughts, but some interviewers will say no. It is a good idea to make a habit of noting risks and next steps as you work through the case so that this step is quicker
- Your answer should:
 - Clearly state the recommendation
 - Support with a few key pieces of evidence (don't rehash analysis – get to the answer)
 - Identify risks
 - Suggest next steps
- Presentation of this should be no longer than 1 min (and sometimes the interviewer will ask for a shorter answer)
- If your interviewer gives you a minute to prepare, you can write out a recommendation slide and present it, similar to the framework at the beginning of the case. This is not necessary or possible in many case situations

Frameworks

Structure/ Framework Building & Explanation

- The most important thing to remember about frameworks is that there is no one-size-fits-all framework.
- That being said, the profit framework applies to a large number of cases; a framework for evaluating an investment decision is also useful.
- For a good overview of how various parts of a framework fits together in one monster diagram, refer to *How To Get Into the Top Consulting Firms* by Tim Darling book. There are good overviews of the most common frameworks in *Case In Point* and in the shared Consulting Club folder, but **do not memorize them**. Instead, you should be able to use your intuitive business judgment to determine the most relevant issues to examine.
- When constructing your framework, **remember to be MECE**, a McKinsey-coined term that stands for Mutually Exclusive and Collectively Exhaustive – meaning that your framework should cover all the key issues, without repeating issues across buckets.

Frameworks

Additional advice

- Rehashing a generic framework like the BCG growth matrix or Porter's Five Forces in a consulting interview basically means interview suicide. Come up with your own framework (may be informed by Five Forces, etc) and make sure to customize the terms for the case: e.g. for a cable company, price is "subscription fee"; for a non-profit, price is "average donation amount".
- Practice structuring the problem by doing "practice starts", where you read the client problem and sketch out a framework. The advantage to this is that you can do it without a case partner.
- Talking through your framework
 - Once you have structured your framework, you will need to walk your interviewer through your thinking. The verbal communication of your framework is just as important as the structure.
 - Make sure you go through in a systematic, logical way, while being concise. Inserting a hypothesis when explaining your framework (eg. "It is my hunch that the profit decline is a result of a potential decline in the demand for razor blades due to a new trend of consumer dislike for shaving") will get you an extra gold star.
 - It is best to explain the framework at a high level by briefly introducing each bucket before drilling down into the issues listed within each bucket.

Math

The Math!

- For some people, doing calculations is the easiest part of the case. For others, it is the most miserable. The key is to be comfortable doing math on pen and paper with someone else watching you, which doesn't always come naturally. That's why it's critical to practice your "public math" through practicing cases with others.
- But keep in mind that 99% of the time, the math will be simple arithmetic, and keeping a cool head will prevent you from making mistakes. Additionally, your interviewer will frequently allow you to round to make the math easier (e.g. 50 week instead of 52)

Sanity checking

- Once you come up with a number, "sanity-check" your answer – that is, make sure it makes sense in the context of the case. If you've sized the US cell phone market to be 10x the US population, you should verbalize to your interviewer that that doesn't seem to make sense and then work back through your calculation.

Mistakes are OK

- You can recover from a math error – as long as you don't let it fluster you. Many of us made math errors in our final rounds of interviews and still received job offers.
- That said, it is best not to make the same mistake more than once, so if you're having trouble with the same issues (e.g. dropping zeros) in practice cases, focus on eliminating those errors.

Math

Talking through the process

- The toughest part about “public math” is learning not to fall completely silent while you’re doing multiplication. You can’t leave your interviewer hanging while you dive into calculations for several minutes, so it’s very important to come up for air and explain what you’re doing.

Use shortcuts

- Remember to use shortcuts – round when you need to, or manage your zeros with scientific notation if that works for you.

So what?

- The point of the case interview is not to showcase your mental acrobatics, but to actually do something with the numbers once you’ve calculated them. One of the most important qualities that interviewers look for is the ability to tie the number back to the point of the question (the answer isn’t “\$8B”, the answer is, “the potential market size is \$8B, which looks pretty attractive ...” and so on). It’s important to not only get the number but to understand how it fits into the overall problem and what it means you should do next.

Practice!

- No one is a natural at all of these steps! It’s important to practice until you’re comfortable doing this in front of an interviewer.

Company Overviews and Interview Processes

Firm Overview

- “Whether it's business strategy, technology strategy or operations strategy, we drive value, shape new businesses and design operating models for the future. That’s high performance, delivered.”

What they look for

- Accenture traditionally recruits on-grounds for its Strategy and Operations, Federal and IT Strategy practices. Accenture looks for well-rounded candidates who demonstrate a passion for creating client value through practical, implementable solutions.

Interview process

- First and second rounds are both on-Grounds, with a week or two between each round
- Second round interviews are likely to be conducted by at least one Managing Director
- Accenture’s interviews are less quantitative
- Interviews exhibit a “conversational style,” where the focus of the case interview is how the candidate communicates his approach and thinking.

Career Hierarchy
Managing Director
Senior Manager
Manager
Consultant (post-MBA)
Analyst

A.T. Kearney or ATK

\ Car-Knee\

ATKearney

Firm Overview

- “A.T. Kearney is a global team of innovative, insightful and collaborative experts who deliver creative, meaningful and, above all, sustainable results. Consultants that generates powerful strategic insights to address practical, real-world needs”

What they look for

- ATK recruits on-grounds at Darden for its North America offices. Given ATK’s national staffing model it makes little difference which office an applicant applies. ATK looks for candidates who are willing to work in its collaborative engagement model.

Interview process

- First round is on-grounds with Principal or Manager
- First round tests quant skills and ability to structure
- Second/Final round is at ATK office
- Second round has three components
 1. Typical case with high-level slant with Partner or Principal
 2. Live case with presentation to panel of Managers and Principals
 3. Behavior fit interview with Partner

Career Hierarchy
Director/ Partner
Principal
Manager
Associate (Post-MBA)
Analyst

Firm Overview

- “Bain & Company is the management consulting firm the world's business leaders come to when they want enduring results. Together, we find value across boundaries, develop insights to act on, and energize teams to sustain success. We're passionate about always doing the right thing for our clients, our people and our communities, even if it isn't easy..”

What they look for

- Generalist practice – no specializations
- Strong emphasis on cultural fit
- Want to see a balance of quantitative ability and qualitative strengths
- “Answer first” structure to responses
- Focus on driving results for the client (originated “fees at risk” model)

Interview process

- First round is at Darden (two cases, with behavioral questions at the start)
- Second round is in the office you are interviewing for; either:
 - 1 typical case, 1 written case, 1 behavioral interview
 - 2 typical cases, 1 written case

Career Hierarchy
Partner
Principal
Manager
Consultant (Post-MBA) → Case Team Leader
Associate Consultant

The Boston Consulting Group



Firm Overview

- “Our customized approach combines deep insight into the dynamics of companies and markets with close collaboration at all levels of the client organization. This ensures that our clients achieve sustainable competitive advantage, build more capable organizations, and secure lasting results.”

What they look for

- BCG looks for intellectually curious candidates that are looking to grow with the firm and are open to new challenges. Fit with the firm’s culture is important, so it is good to start developing relationships with BCGers early on in the process. There is no set professional background or profile that BCG is looking for in prospective Consultants.

Interview process

- First round is at Darden (two cases)
- Second round (two cases) is in the office you are interviewing for; international offices do second round in the US
- All cases likely to have quantitative component, especially first round
- The second round weighs “fit” with the office/firm more heavily than the first

Career Hierarchy
Partner
Principal
Project Leader
Consultant (Post-MBA)
Associate

Firm Overview

- “As the world’s largest management consulting firm, we help organizations build value by uncovering insights that create new futures and doing the hard work to improve performance.”

What they look for

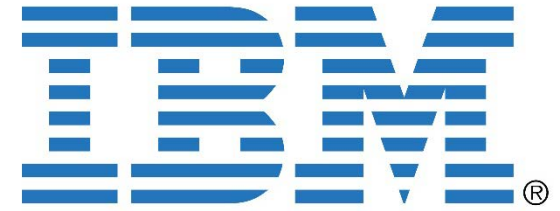
- Deloitte recruits on-grounds at Darden its Commercial Strategy & Operations practice, although Technology and Human Capital are also a part of the consulting division. They look for well-rounded candidates who understand their mantra of “executable strategy” – Deloitte prides itself on not only doing the strategy assessment, but staying on for the implementation and execution of the strategy as well.

Interview process

- First and second round are back-to-back at Darden
- The first round case will heavily test quantitative ability
- The second round will have a group case as well as a more traditional case
- Deloitte interviews often feature a “data sheet” that presents all the information in advance, but it is still important to follow the standard case format

Career Hierarchy
Director/ Principal/ Partner
Senior Manager
Manager
Senior Consultant (post-MBA)
Consultant
Business Analyst

IBM Global Business Services



Firm Overview

- In 2011, Fortune ranked IBM the 18th largest firm in the U.S and the 7th most profitable. GBS consultants help their clients achieve competitive advantage by framing industry opportunities and challenges into actionable strategy

What they look for

- IBM GBS recruits at Darden for its Senior Strategy & Analytics Consultant position. They look for well-rounded candidates with excellent analytical and communication skills.

Interview process

- Internship: 1 round of 3 interviews total, with 2 case interviews and 1 behavioral interview
- Full-time: 2 rounds of interviews
- Interview formats can deviate from the above depending on the IBM GBS group interviewing (eg. public sector, commercial)

Career Hierarchy
Partner
Associate Partner
Senior Manager
Manager
Senior Consultant (post-MBA)
Consultant

Firm Overview

- “McKinsey & Company is a global management consulting firm. We are the trusted advisor to the world's leading businesses, governments, and institutions.”

What they look for

- McKinsey looks for five aspects in its candidates. All areas are tested in the interview process; the case interview will test problem solving and the other skills will be tested through the experience interview. They are **Personal Impact, Entrepreneurial Drive, Problem-Solving Skills, Achievement** and **Leadership Abilities**

Interview process

- First round (on-grounds): 2 interviews, each with 1 case and 1 experience question
- Second round (in-office): Generally 3 interviews, each with 1 case and 1 experience question
- First rounds normally conducted by EMs; second rounds by APs/ Partners

Career Hierarchy
Partner
Associate Principal
Engagement Manager
Associate (Post-MBA)
Business Analyst

The Parthenon Group



Firm Overview

- The Parthenon Group is a strategy consulting firm with expertise in private equity and education. Additional practices include healthcare, industrials, and consumer and retail. Parthenon operates a “two-case model” and typically emphasizes light-travel. Parthenon was acquired by EY in September of 2014, but is maintaining a separate brand and operations.

What they look for

- Parthenon recruits at Darden for its Consultant and Summer Consultant roles. Parthenon describes its consultants with the phrase “Smart. Nice. Driven.”

Interview process

- Format is the same for summer and full-time roles
- First Round: 2 behavioral and 1 case interview
- Second Round:
 - 15 minute chat with a Managing Director
 - 1 hour case (30 minutes solo preparation, 30 minutes talking through the case with a vice president or managing director)

Career Hierarchy
Managing Director
Vice President
Sr. Consultant
Consultant (Post-MBA)
Senior Associate
Associate

Firm Overview

- “From expansion into emerging markets, to new technologies such as cloud computing, companies are trying to sort through how to exploit new opportunities for growth. We have deep industry experience and deliver customer solutions to help our clients address their most complex business issues. Our professionals bring a diverse background of skills and education allowing us to provide clients with a unique and holistic perspective on their issues.”

What they look for

- Fit, and the typical consulting analytical toolkit, as well as relevant background experience. PwC is functionally-based, with no one entering as a generalist.

Interview process

- Round 1 (on-Grounds): 2 behavioral interviews, focusing on fit and relevant background experience
- Round 2 (in-office): 2 behavioral + 1 case, and 30 min for additional questions
 - The case background will be distributed 48 hrs in advance for review; the questions will be given out in the interview

Career Hierarchy
Partner
Director
Manager
Senior Associate (post-MBA)
Experienced Associate
Associate

Firm Overview

- A mid-sized management consulting firm focused on sales and marketing, primarily in healthcare

What they look for

- ZS recruits at Darden for its Consultant position. The firm values candidates with strong quantitative abilities and marketing knowledge. A thorough understanding of the pharmaceutical and medical devices industries is also valued.

Interview process

- First round interview held on-grounds at Darden
- Second round typically held off-grounds at the Princeton office
- Quantitative skills and ability to synthesize large amounts of information are rigorously tested

Career Hierarchy
Principal
Associate Principal
Manager
Consultant (post-MBA)
Associate Consultant
Analyst

Practice Cases

Case 1:

Call Center Conundrums



Firm Style	Interview Round
Bain	1

Case Question

We have been contracted by a client who operates five domestic call centers. The centers are spread out in a wide variety of places throughout the United States. The company is considering consolidating their offices into a single location. How would you advise them?

Clarifying information *(provide only if corresponding questions are asked)*

This case prompt is intentionally open and ambiguous. There are questions to allow the interviewer to lead the case if necessary, but this should be interviewee-driven

Where are the call centers?	Domestic. No further information is available
Would the client like to expand internationally?	They are open to all suggestions
How many employees does the client have?	500, equally distributed across locations
Any other question	Will get into that later in the case

Call Center Conundrums

- Framework/ Structure

- WEAK: mentions revenue and cost, some general thoughts on how profits are generated
- GOOD: clearly breaks out revenues and costs
 - Revenues
 - Price – Cost per call or set contract. Do they charge different rates at different locations
 - Volume – Calls per day. Excess capacity. How calls are dispersed across locations
 - Costs
 - Variable – labor, utilities, phone bills, etc
 - Fixed – cost of offices, SG&A, insurance, etc
- GREAT: identifies that this question is a before/ after analysis to determine attractiveness of move
 - Costs before – 5 offices, # employees, SG&A
 - Costs after – 1 office, # employees, presumably lower SG&A
 - Any impact on revenue

- Can also prompt interviewee to consider:

- One time zone v. multiple time zones; other factors affecting quality of service available
- Managing layoffs

Call Center Conundrums

- Question 1: The client has chosen a potential location for the consolidated center just outside Detroit. What costs should they expect if they relocate to this location?
 - Interviewer – provide numerical data (below) to support the costs *after the interviewee identifies them*. Do not prompt calculations, but can prompt cost identification with leading questions like “Is it reasonable to expect every employee to move to Detroit?”

Cost information

- **Labor relocation:** Client has offered \$5K/employee to relocate
 - Client has 500 employees
- **Severance:** Client expects to lose 50% of employees due to reluctance to move and will pay them 6 weeks’ salary as severance. Average salary is \$78K
- **Office space:** New office is equal to total cost of original 5 offices
- **New employees:** Acquisition and training of new employees will cost \$3K/ employee
- **Cost of physical move:** Negligible – new office is fully equipped

Math:

- **Labor relocation:** $500 \text{ employees} * 50\% = 250$;
 $250 * \$5K = \$1.25M$
- **Severance:** $\$78K \text{ salary} / 52 \text{ wks} = \$1.5K/\text{wk}$ *
 $6 \text{ wks} = 9K * 250 \text{ employees} = 2.25M$
- **Office space:** \$0 net cost
- **New employees:** $250 \text{ employees} * \$3K = \$0.75M$

Total: \$4.25M

Call Center Conundrums

- Question 1 Insights

- Stronger interviewees will generate insights without prompting
- Relocation costs \$4.25M
- Relocating all employees would save \$1.75M on the total cost, since it costs \$12K per employee to hire new employees (severance + acquisition/ training) and \$5K to relocate. Should therefore encourage employees to make this move
- However, given the \$4.25M cost (lower bound of \$2.5M if all employees move), would like to see other factors in favor of relocation – e.g. additional savings
 - This should lead into Q2. If interviewee does not identify that additional factors must be present to decide to make this move, prompt with next question

Call Center Conundrums

- Question 2: What would need to be true for this to be an attractive place for the client to relocate?
 - Interviewee should brainstorm a list of things that could make this appealing – a non-exhaustive list is:
 - More contracts
 - Lower wages
 - Lower SG&A
 - Better tax benefits
 - Proximity to customers
 - No danger of losing customers or decreasing customer service due to consolidating time zones
 - Lower cost of living for employees
 - Strong interviewees should ask whether there is information about any of these potential savings – if not, prompt with Q3

Call Center Conundrums

- Question 3: We have identified two advantages to relocating. First, because the new location does have a significantly lower cost of living, labor costs will decrease by 5%. Second, the client will be able to reduce their SG&A budget by 20%

Information to provide (if asked)

- For simplicity, the \$78K annual salary represents all-in labor costs (salary + benefits)
- SG&A budget: \$10M annually
- interviewee should not have to ask for number of employees (provided in previous question – 500)

Math:

- **Salary savings:** $\$78K * 5\% * 500 \text{ employees} = 1.95M$ (can round to 2M)
- **SG&A savings:** $10M * 20\% = 2M$

Total annual savings from move: \$4M

• Insights:

- The interviewee should recognize that the client can expect to recoup almost the entire cost of relocation (4M of 4.25M) within the first year, making the move very attractive
- If the interviewee suggests a NPV calculation, tell them that the client evaluates decisions like this with a 5-yr payback period
- If the interviewee dives into the risks of paying new employees a lower salary and/ or decreasing salary of current employees, explain that the 5% decrease actually comes from a decrease in cost of benefits to the company (this is a good thing for the interviewee to recognize, but don't discuss it further)

Call Center Conundrums

- Question 4: The partner on the case has asked you what he should tell the client's CEO on a call in 2 minutes
 - A strong interviewee will provide the recommendation to move in the first sentence, support with the compelling cost-savings information and acknowledge some of the risks
 - Insights/ risks could include:
 - Cheaper labor and SG&A savings make Detroit an attractive place to consolidate the call center business. With a one-time up-front relocation cost of \$4.25M, could save \$4M annually
 - The costs hinges on a significant number of employees moving to Detroit, which may not be realistic. If fewer than 50% of employees agreed to relocate, severance costs would increase. Conversely, this would be even more attractive if more than 50% of employees relocated. Next step to look into factors that could influence them to make the move
 - Would want to ensure that the move would not impact the quality of service experienced by current customers – this analysis assumed that revenues remained unchanged
 - Any unanticipated costs could also impact this analysis – would like to be more confident in these numbers

Case 2:

Blood Banking



Firm Style	Interview Round
BCG/ Bain	2

Case Question

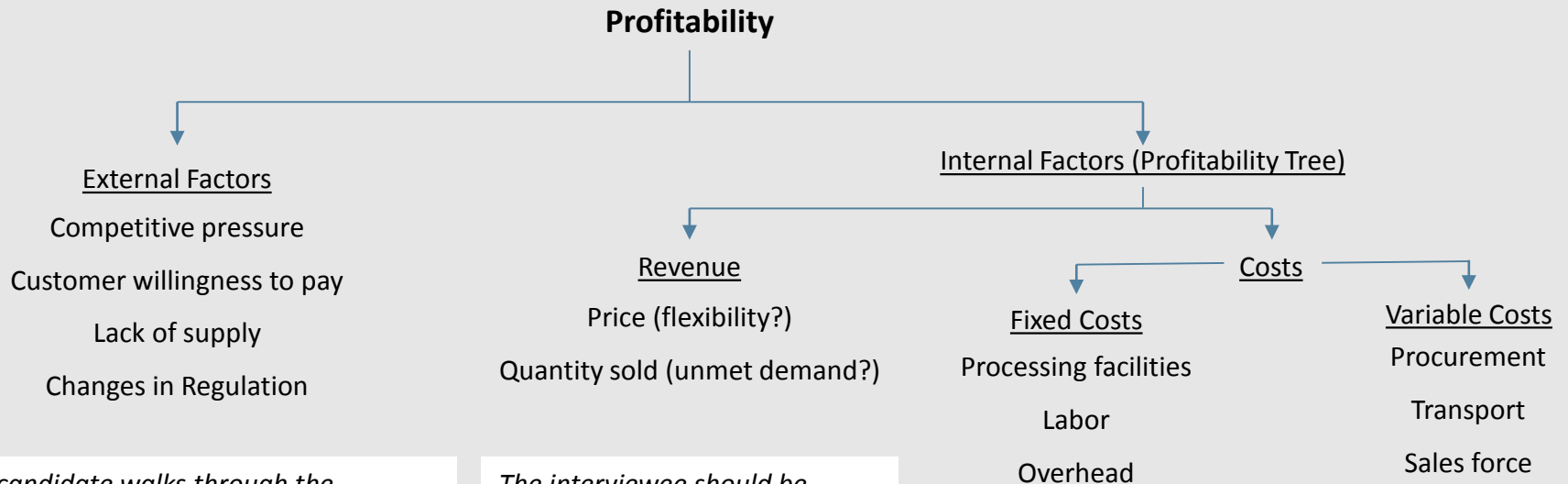
Our client is a major blood bank with operations in four southern states. They collect blood through drives at locations like schools and offices. The blood bank has one processing center in each state where the blood is sent for testing and treatment. The blood is then transported to hospitals to be used. Only around 80% of the demand for blood is met, and hospitals often share blood among themselves although transportation is expensive. Our client has been losing money for some time. The CEO has called us in to understand how they can improve profitability and prioritize their next steps.

Clarifying information *(provide only if corresponding questions are asked)*

Does the client face competition in its markets?	Yes, in each state where it operates there are other blood banks
How long has the client been experiencing losses?	For several years
How great are the losses the client is facing?	No information available
Are there any substitutes for human blood?	No

Blood Banking

Recommended Framework



*As the candidate walks through the framework, the interviewer should tell the interviewee that **there have been changes in regulation that require the blood to now be purified to a higher degree. These will come into effect at the beginning of the next fiscal year.***

*The interviewee should be probed to get insights on competitive pressure, price elasticity of demand for blood and the shortage of blood supply (see next slide). Interviewer should mention there is no possibility of increasing supply **AFTER** the interviewee explores the possibility of increasing quantity sold.*

The interviewer should steer the interviewee away from discussing cost cutting, since the client has no scope to cut costs further.

Blood Banking

- Key insights from framework

- **Competition:** Even though the blood bank faces competition, the interviewee should recognize that because only 80% of the demand for blood is met, competition cannot be causing price wars. The interviewee should be told that we have no further information about the competitors.
- **Customer willingness to pay (price elasticity):** Blood is in high demand and short supply, therefore demand is not price-sensitive.
- **Supply of blood:** Ideally the interviewee should brainstorm ways to increase the supply of blood – e.g. raising awareness, increasing frequency, etc. The interviewer should encourage brainstorming, and then tell the interviewee that blood supply is fixed.
- **Regulations:** The blood donation process is heavily regulated and the interviewee should identify regulatory changes as a potential external factor that could affect profitability. **The interviewer should prompt the interviewee if they do not identify regulatory issues independently.** They should then be told about the regulatory changes requiring more purification; a strong interviewee will recognize that this impacts all competitors and is putting upward pressure on processing costs.
- **Costs:** Interviewee expected to propose costs as a potential way to increase profit while talking through the framework. Interviewer should share that there is no scope for cost-reduction.
- **Path forward:** Interviewee should propose exploring the profit side of the profitability tree, as there is no scope for cost-reduction. The interviewer should then ask them to first take a step back and consider the current profitability situation (Q1 – next slide).

Blood Banking

- Question 1: How do we know if the client is breaking even, and what is the extent of their current losses?

Information to be provided (if asked):

Fixed costs: \$15M annually

Variable costs: \$80/ unit

Units sold: 400K

Price per unit: \$100

Math:

Breakeven:

Number of units sold = 400K

Contribution per unit: $\$100 - \$80 = \$20/\text{unit}$

Units needed to break even: $\$15\text{M fixed costs} / \20
contribution/ unit = **750K units annually**

Gap to breakeven: 750K units needed – 400K sold = 350K

Current financial situation:

400K units * \$20 contribution = \$8M

\$8M contribution - \$15M fixed costs = **(\$7M) loss annually**

• Insights:

- The client is selling 350K fewer units than necessary to break even, and is losing \$7M annually
- The interviewee should realize that, as there is no scope to reduce costs or increase blood supply, they should evaluate whether prices can be increased

Blood Banking

- Question 2: How much does the client need to increase the price to break even?

Math:

Number of units sold = 400K

Per-unit contribution needed to break even:

$\$15\text{M fixed costs} / 400\text{K units sold} = \37.50

Current contribution \$20

Additional contribution needed: $\$37.50 - \$20 = \$17.50$

- Insights:
 - The client can break even at 400K units sold annually with an **increase in contribution of \$17.50 per unit, to \$37.50 contribution in total**. If we can assume the same costs, this would mean an **increase in price to \$117.50 per unit**
 - The interviewee should then move towards discussing whether a 17.5% price increase is feasible, and request information about customer willingness to pay or overall hospital costs

Blood Banking

- Question 3: Are hospitals likely to pay the increased price of \$117.50?

Information to be provided (if asked):

- Hospitals use 2.5 units of blood per surgery. Assume no other usage of blood
- Total cost of an average surgery is \$40K

Math:

Current blood cost to hospital per surgery: 2.5 units * \$100/ unit = \$250

% of blood cost per surgery = \$250 blood cost/ \$40K total cost = 0.625% (can round to <1%)

- Insights:

- Blood supply represents a very small percentage of total cost of surgery; hospitals should not be severely impacted by the marginal price increase
- Furthermore, given the limited supply, there is no danger of losing customers to competition due to a price increase

Blood Banking

- Question 4: Once the new regulation requiring a higher degree of purification comes into effect, what does the client need to do to break even?
 - The interviewee should have identified earlier that the new regulations will result in an increase in variable costs and potentially fixed costs (new equipment for purification, etc) – if they do not suggest discussing how to change price as a result, point them in this direction

Information to be provided (if asked):

Fixed costs: Same

Variable costs: 10% increase

Units sold: Same

Math:

Number of units sold = 400K

Contribution per unit to break even = \$37.5 (\$15M/400K)

- From Q2 calculation

New Variable costs = \$88 (110%*\$80)

Price per unit to break even = \$125.5 (\$8 in additional variable costs + \$37.5 contribution)

• Insights

- The client needs to raise prices by 25.5% to \$125.50/ unit to break even once the new regulations come into effect
- This should be feasible given the previous analysis of the cost of blood as compared to overall operating costs. Additionally, all competitors will experience similar increases in costs. The sales team can explain the price increase is necessitated by the new regulations

Blood Banking

- Question 5: Can you wrap up with a recommendation to the CEO?
 - Recommendation should include:
 - Price of blood needs to be raised by 25.5% to \$125.50/ unit to break even once the new regulation kicks in
 - Potentially there is room for increasing the price even more, given the price inelasticity and unmet demand, which would allow the client to make a profit
 - Risks to consider:
 - No serious risk to reasonable price increases given that blood is in short supply
 - Reputational damage if prices are raised significantly out of line with competition
 - Next steps:
 - An immediate increase of 17.5% could prevent further losses; should determine a target for profit margin after the new regulations take effect and set the appropriate pricing strategy

Case 3:

Lonely Gas Station



Firm Style	Interview Round
BCG	Final

Case Question

Our client is the owner of a gas station between towns A and B – 10 miles to each town. He is wondering if it would make sense to add a convenience store to the gas station

This is intentionally open-ended. The interviewee will have to ask multiple questions to frame the problem and gather the information necessary

Clarifying information *(provide only if corresponding questions are asked)*

Does the client have any competitors?	There are no other gas stations in town A or B
Does the gas station currently sell anything besides gas?	Gas is 75% of revenue (10% profit margin) and the gas station also offers car washes and other services (25% of revenue, 20% profit margin)
What would the client like to consider to decide if it “makes sense”?	• Making profit • Having a better chance to hold off new competition entering the market • Diversifying income
Who are the gas station’s current customers?	Customers are from towns A and B; there are no other customers

Lonely Gas Station

- Framework could include:
 - Revenue/ profit of current and potential business
 - Gas station business
 - Convenience store business
 - Potential revenues, costs, initial investment cost
 - Would adding the convenience store bring in additional gas station customers
 - Market/ competition
 - Market size
 - Other convenience stores?
 - Growth in town A and/ or B?
 - Changes in preferences for gas (electric cars? Public transportation?)
 - Product and client capabilities
 - Knowledge of retail/ convenience stores
 - Supplier contacts
 - Management experience; potential to contract an external party to run convenience store

Lonely Gas Station

- Question 1: How profitable is the current business?
 - The interviewee should structure a logical approach to market sizing on their own; provide these numbers only after they identify that they need the information

Information on market:

1000 people in each town

80% of population owns a car

50% of car-owners get gas from client

Math:

Per town: $1000 * 80\% = 800$; $800 * 50\% = 400$

Total: $400 * 2 = 800$ person customer base

Information on profit:

How often do our customers get gas? Once/ wk

- Can tell interviewee to round down to 50 weeks/ yr

What do they spend each time? \$50

Do our customers make all their gas purchases at our station? No, only 40%

Interviewee should have already asked about gas station revenue (previous slide) and learned that gas makes up 75% of revenue (10% profit margin) and other purchases are 25% (20% profit margin). **Prompt if not: Do you think the entire purchase is gas each time?**

Math:

$800 \text{ customers} * \$50 * 50 \text{ weeks/ year} = 2\text{M on gas per year}$

$2\text{M} * 40\% \text{ of purchases made at client} = \$800\text{K client revenue}$

75% of revenue is gas: $\$800\text{K} * 75\% = \$600\text{K} * 10\% \text{ profit margin} = \60K profit

25% of revenue is other: $\$800\text{K} * 25\% = \$200\text{K} * 20\% \text{ profit margin} = \40K profit

$\$60\text{K} + \$40\text{K} = \$100\text{K profit per year}$

Lonely Gas Station

- Question 2: How profitable would the convenience store be on an ongoing basis?

Information on convenience store:

Gasoline customers will spend an additional \$20 at the convenience store per purchase, but will not increase frequency of purchases

Recurring costs for convenience store are:

- Labor: \$75K/ yr
- Utilities: \$5K/ month
- COGS: 50% of revenue

ONLY PROVIDE IF ASKED: 50% of *town population* who currently are not customers (the remaining 1200 non-customers, not just the 80% car owners) will spend \$5 per week at the convenience store

Math:

Revenue – existing customers:

- $800 \text{ customers} * 50 \text{ weeks} * 1 \text{ purchase/ week} * 40\% \text{ purchases made at our gas station} = 16\text{K total purchases}$
- $16\text{K purchases} * \$20 = \320K addl revenue

Revenue – new customers:

- $1200 \text{ non-customers} * 50\% * \$5 \text{ per week} * 50 \text{ weeks} = \150K
- **Total revenue: $\$320 + \$150 = \$470\text{K}$**

Costs

- $\text{COGS: } \$470\text{K revenue} * 50\% = \235K
- Labor: \$75K
- Utilities: $12 \text{ months} * \$5\text{K} = \60K

Profit:

- $\$470\text{K} - \$235\text{K} - \$75\text{K} - \$60\text{K} = \textbf{\$100K per year}$

Lonely Gas Station

- Question 2b: Given that, what is the NPV of opening a gas station?
 - Interviewee should ask for:
 - Up-front investment cost: \$1M
 - Client's discount rate: 10%
 - Town population growth rate: 0%
 - Expected life: perpetuity
 - Calculation:
 - $\$100\text{K annual profit} / 10\% \text{ discount rate} = \1M
 - Investment cost of \$1M
 - $\text{NPV} = 0$

Lonely Gas Station

- Question 2c: What does the NPV of 0 mean for our client?
 - A strong interviewee would have moved to these insights after calculating that $NPV = 0$; if not, prompt with this question
 - The interviewee can provide a recommendation either for or against the convenience store as long as it recognizes the NPV of 0 and is backed up by logic
 - For investment: Diversification; gain new customers who may start purchasing gas/ potentially increase frequency of visits of current customers to store
 - Against investment: NPV of 0; could be even worse in reality; high investment cost

Lonely Gas Station

- Question 2d: What happens to the NPV if a convenience store competitor enters the market?
 - The interviewee should recognize that a competitor entering the market would make the situation less attractive for our client
 - Should also mention that a competitor likely has a 0 NPV at best given our client is an established brand in the market with an established customer base, so it seems unlikely that a competitor would want to enter
 - Strong interviewees could take initiative to discuss ways to improve NPV if a competitor enters the market

Lonely Gas Station

- Question 3: The client also asked us to provide him with a list of things to consider in this decision. What would you tell him?
 - Looking for a qualitative discussion of things to consider; a balanced answer will indicate both potential benefits and threats to widening the client's business
 - The interviewee can ask for a minute to brainstorm; a strong interviewee will have a structured approach
 - Answer could include:
 - Current customers: higher convenience to them, higher revenue, increasing frequency of visit, longer wait times
 - New customers: can they be converted to buy gasoline, is there potential to attract customers from other cities
 - Competition: new entrants, supermarkets/ restaurants/ other similar offerings, could the convenience store differentiate with weekend/ night hours or something similar
 - Other: is there enough space to add a store? Are there any regulatory issues?

Case 4:

Mutual Funds



Firm Style	Interview Round
BCG	Final

Case Question

A large retail bank, Retail Bank Co, is currently #1 or #2 in nearly all major retail segments, including deposits, loans and mortgages. The bank's mutual fund business is currently far behind its competitors, ranked #5. They have hired BCG to determine why the mutual fund business is lagging behind.

Clarifying information *(provide only if corresponding questions are asked)*

What is the client's primary objective?	To achieve top-line revenue growth
How are mutual funds priced?	Percentage commission on assets under management (AUM)
Does the mutual fund business offer multiple products?	Just one mutual fund product, in line with market offerings

Background info for interviewer

This case is meant to be relatively tough and the interviewer should provide little help or guidance. **The key issue is that the incentives for the retail branch employees are driving them to sell products other than mutual funds.** Until the interviewee identifies this, shoot down every hypothesis posed – more on this on 2nd slide

Mutual Funds

- How to work through this case:
 - Interviewee should develop a framework (slide 3) and choose an area to start with, looking for revenue drivers that can be changed or that could be contributing to the lagging performance
 - Should understand that the components of revenue are the commission charged and the total assets under management – so, # customers, amount invested per customer – and be looking for ways to increase those 3 things
 - For every driver/ hypothesis, interviewer should either tell interviewee that client is the same as competitors (e.g. brand recognition/ reputation, product performance, etc) or that it cannot be changed (e.g. commission is in line with market and can't be increased) – whichever makes sense
 - **Until the interviewee asks about salesforce incentives structure – then provide the information on slide 4**
 - If they ask about sales more generally, ask them to be specific about what drives sales
 - Then move the interviewee to calculating revenue (slide 5)
 - Finally, ask for a summary recommendation (slide 6)

Mutual Funds

- Framework could include
 - Customers
 - Number of customers, growth
 - Amount invested per customer
 - Willingness to pay (price, commission)
 - Product demand (mix, maybe customer preferences have changed/ they want a different product)
 - Product
 - Performance
 - Benefits to the buyer
 - Substitutes
 - Channels
 - Retail Bank Co
 - Capabilities (are employees worse at selling mutual funds? Is client worse at managing mutual funds?)
 - Intangibles (brand)
 - Competitors
 - Value proposition (price for performance)
 - Performance
 - Brand

Mutual Funds

- Information to provide on salesforce, when corresponding questions are asked
 - Sales employees are paid a bonus once they reach \$10M in total sales
 - They sell two products, mortgages and mutual funds
 - Mortgages average \$250-\$500K apiece
 - Mutual funds average \$10K invested per customer
 - A sales employee spends about twice as much time with a mortgage customer as a mutual fund customer
 - Nothing is known about incentives structures in the client's competitors
- Insights
 - Interviewee should understand that retail employees have been incentivized to sell mortgages, rather than mutual funds, because their size makes it easier to hit the bonus threshold and they only take twice as long to sell
 - Should generate suggestions to improve this – anything is acceptable as long as it is logical and would create more incentive for the client's sales employees to sell mutual funds

Mutual Funds

- Question: What is the current revenue from the mutual funds product line, and how would profits increase if we doubled the client's assets under management (AUM)?
 - A strong interviewee should articulate that doubling AUM could come from growth in customers or growth in amount invested per customer

Information to be provided (if asked):

Price – commission of 1% on AUM

Volume

- \$10K invested per customer
- 2M mutual funds customers

Current costs:

- Operating margin: 75%
- Fixed costs: \$50M

Changes to costs:

- Doubling the amount invested per customer would not impact costs
- Doubling the number of customers would require increasing our customer service staff, bringing fixed costs to \$90M

Math:

Total AUM = $2M * \$10K = \$20B$

Total revenue = $1% * \$20B = \$200M$

Current profit:

- $\$200M * 75\% = \$150M$ operating profit
- $\$150M - \$50M$ fixed costs = **\$100M**

Profit if double customers:

- $\$400M * 75\% = \$300M$ operating profit
- $\$300M - \$90M = \mathbf{\$210M}$

Profit if double amount invested:

- $\$400M * 75\% = \$300M$ operating profit
- $\$300M - \$50M = \mathbf{\$250M}$

Doubling AUM increases profits by 110-150% - more beneficial to increase amount invested per customer than to get new customers at same average investment

Mutual Funds

- Question: Please sum up your work into a brief recommendation we can share with the client
 - Recommendation: Adjust salesforce incentives structure to encourage sales force to sell mutual funds, increasing mutual fund AUM and improving position against competition
 - Currently have very similar performance and capabilities to competitors, but sales force bonus structure encourages them to focus on other products
 - Doubling amount invested per customer could increase profits by 150%
 - Risks:
 - Increasing incentives for selling mutual funds will divert salesforce attention from other business lines – could potentially impact mortgage business
 - Could be challenging to manage rapid growth in AUM
 - Next steps:
 - Consider dedicating a sales force to mutual funds
 - Find ways to differentiate from competition based on performance, customer service, etc, to be able to charge a higher price

Case 5:

Whale Hotel



Firm Style	Interview Round
McKinsey & Co.	1

Case Question

Our client is a real estate company that owns and operates luxury hotels around the world. They've previously owned 3 resorts in Dubai and are considering building a fourth, targeted specifically at high net worth individuals they refer to as "Whales".

Clarifying information *(provide only if corresponding questions are asked)*

What is the investor's payback period?	5 years
How long is the construction process?	2 years
Can you describe the tourism industry in Dubai?	Seasonal (25% increase in summer)
Does the company currently own any hotels in Dubai?	No – ignore cannibalization
Are there any similar resorts in Dubai currently?	Yes – the King's Palace, the Egyptian and the Belzor

Whale Hotel

- Question 1: After looking at **Exhibit 1**, calculate total number of whales expected in these hotels per night:
 - If necessary, prompt interviewee to do this calculation for peak and off-peak nights separately

	Whales Per Night, Off-Peak	Whales Per Night, Peak
King's Palace	$5000 * 80\% * 25\% = 1000$	$5000 * 80\% * 25\% = 1000$
Egyptian	$4000 * 75\% * 25\% = 750$	$4000 * 95\% * 37.5\% = 1425$
Belzor	$8000 * 75\% * 33.3\% = 2000$ <i>(should see that 33% roughly equals 1/3 and round to 2000)</i>	$8000 * 100\% * 50\% = 4000$
Total	3750	6425

- Interviewee should identify
 - Correlation between % whales and swimming pool quality
 - Impact of seasonality on overall demand (particularly during peak season when the Belzor reaches 100% utilization suggesting there could be additional demand not being met)
- Steer the interviewee towards coming to the conclusion that the **hotel design must include an excellent quality pool** if they do not independently reach this insight
- Ask the interviewee what a **reasonable rate** to charge a whale to stay in the new hotel would be.
 - Any proposed rate with a logical explanation is acceptable. After discussion of rate state that the client is going with a conservative approach of **\$2500/ night** to compete with the Belzor)

Whale Hotel

- Question 2: Market research indicates a hotel with an excellent quality pool charging \$2500 would capture 100% of the whales who stay at the King's Palace and the Egyptian as well as an additional 250 whales per night during the off-peak season and 1575 whales per night during the peak season. What is the expected whale demand for the potential new hotel?
 - Nightly Whale Demand, Off-Peak = 2000
 - $1000 \text{ (King's Palace)} + 750 \text{ (Egyptian)} + 250 = 2000$
 - Nightly Whale Demand, Peak = 4000
 - $1000 \text{ (KP)} + 1425 \text{ (E)} + 1575 = 4000$
- Question 3: After looking at Exhibit 2, how many stories will the new hotel require?
 - Interviewee should recognize hotel will need capacity of at least 4000 rooms to satisfy peak demand
 - With 1500 rooms per story, requires a **3-story hotel (4500 total rooms)**

Whale Hotel

- Question 4: Annual operating expenses are \$250M per 1500 rooms. Land acquisition is \$2B. What is total initial investment and would you recommend the client go forward with the hotel?

Initial Investment

Land - \$2 B

Rooms - \$ 3 B (3 stories)

Pool – \$ 0.5 B (excellent)

\$5.5 B total

Annual Revenue

\$2500/night * 30 nights = \$75000 per whale monthly

4000 whales * \$75000 = \$300 M x 3 months = \$900 M peak

2000 whales * \$75000 = \$150 M x 9 months = \$1.35 B off-peak

Total annual revenue = \$2.25 B

Annual Operating Expenses

(\$250 M per 1500 rooms) x 3 = **\$750 M** for 4500 rooms

Annual Operating Profit

\$2.25B - \$750 M = **\$1.50 B** annually

When interviewee asks, share that the client's investment goal is a 5-yr payback period (should be used instead of a discount rate/ PV calculation)

Payback period

- Client goal: 5-yr payback period
- 2 years of construction: 0
- 3 years of operation: $3 * \$1.5B = \$4.5B$
- After 5 years: $(0 + \$4.5B) - \$5.5B \text{ initial investment} = -\$1B$

- Interviewee should calculate the up-front investment of \$5.5B and then compare the annual operating profit with the investment to conclude the client's investment goal of a 5-yr payback goal will not be met. (Payback period: ~5.67 years)
- Strong candidates will recognize that we have not explored other revenue-generating opportunities for the hotel and suggest ways that this calculation could be made more attractive (restaurant, bar, room service, etc)

Whale Hotel: Exhibit 1

Resort	Number of Rooms	Off-Peak Utilization	Off-Peak % Whales	Rate	Swimming Pool	Peak* Utilization	Peak* % Whales
King's Palace	5000	80%	25%	\$2000/night	Poor	80%	25%
Egyptian	4000	75%	25%	\$2000/night	Good	95%	37.5%
Belzor	8000	75%	33.3%	\$2500/night	Excellent	100%	50%

Note: Peak season runs from June-August

Whale Hotel: Exhibit 2



Variable A: Hotel Stories

1500 rooms per story

\$1 billion construction cost per story

Variable B: Pool Options

Excellent - \$500 million

Good - \$100 million

Poor - \$50 million

Case 6:

BB's Contact Lenses



Firm Style	Interview Round
McKinsey	1

Case Question

Your client is a well-known contact lens provider called BB. BB manufactures and distributes contact lenses in the U.S. BB is one of the largest players in the US market, and has been for quite some time, however, the company feels that compared to its main competitor, it is not doing as well as it could. BB had called in McKinsey to find out how to solve this problem and to recommend a solution.

Clarifying information *(provide only if corresponding questions are asked)*

How is the client measuring its performance/ why does the client think they are “not doing as well as they could”	BB completed a benchmark study of itself v. its biggest competitor and discovered its profits were not growing as fast as the competition
Who are BB's competitors?	There are two main players in the market, BB and Competitor A. They split the market equally. Other, smaller firms exist, but are small enough to be negligible
Does BB operate only in the US?	Yes
Is BB comparable to Competitor A in size, markets covered, products, etc?	Yes

BB's Contact Lenses

- Note for interviewer – How to begin this case:
 - Interviewee should develop a framework (Q1a; possible areas to include on next two slides) and begin working through potential drivers that could be causing the issue (Q1b; info for interviewer to share on next two slides)
 - As interviewee explains how each driver could be causing the issue, allow them to talk you through their thought process, and then provide the information given on the next two slides, explaining why that isn't a viable solution
 - **Except** when the interviewee discusses distribution channels – then move to Q2 and share Exhibit 1
 - If the interviewee does not propose customer mix/ distribution, keep prompting on revenue drivers until they get there
- Question 1a: How would you go about structuring an analysis of this problem?
 - Any logical framework is acceptable; profitability framework works well. Key areas to include listed on next two slides
- Question 1b: Based on the information I've given you so far, and based on what you know about the contact lens industry, where do you think the problem lies?
 - Answer doesn't matter, but interviewee should propose logical hypotheses. As they are discarded (see interviewer instructions above) interviewee should continue proposing ideas –
 - A good way for the interviewee to do this is to state that they are thinking of a current, established major player (i.e. Bausch & Lomb, J&J) and talk through how each driver could play out

BB's Contact Lenses

- Cost Side

- Variable Costs

- **Raw Material:** inputs will be plastic, saline solution (water, salt), packaging (paper, aluminum foil, plastic). These are all commodities, so any issues BB has with raw material costs are likely also experienced by the Competitor. **Not a potential solution**
 - **Labor:** is unskilled, wage rate set by minimum wage standards. Nothing here to put BB at disadvantage to Competitor. **Not a potential solution**

- Fixed Costs

- **Plant, Property & Equipment:** No difference between BB and Competitor in plant costs or plant efficiency. **Not a potential solution**
 - **R&D:** Probably equal between BB and Competitor. **Not a potential solution**
 - **Overhead:** No major differences. **Not a potential solution**
 - **Marketing/Distribution:** No major differences. **Not a potential solution**
 - **Legal issues:** None currently, and BB is probably big enough that even a huge class action settlement shouldn't affect its bottom line too much. **Not a potential solution**

BB's Contact Lenses

- Revenue Side

- Pricing

- Contact lenses are fairly commoditized. Minor differences in pricing may exist, but probably nothing major overall. Customers may be price sensitive, but given that lenses are fitted to a person by their doctor, customers do not purchase lens purely on price. Comfort/fit and compatibility are considerations, for example.

Not a potential solution

- Volume

- Substituting away from BB with Lasik, glasses, etc: Would hit Competitor equally.

Not a potential solution

- Market share: BB and Competitor have equal market shares, which has been fairly constant even with significant advertising efforts from both sides. Client would like the team to figure out why BB is less profitable at this market share.

Not a potential solution

- **Mix – when interviewee gets to the drivers below, push them to explain why it would be relevant, and then move to Q2**

- Product mix: Maybe Competitor is selling more profitable mix of lenses?
 - Distribution mix: Maybe Competitor is selling through more profitable channels?

BB's Contact Lenses

- Question 2: McKinsey analyzed the distribution channels of BB and its competitor, and came up with the following information (show Exhibit 1: Customer Mix slide)
 - Slide is intentionally vague and there is no specific question asked. Interviewee should proactively talk through the slide, asking clarifying questions as necessary
 - Slide shows customer mix in terms of volume. Interviewees should remember that BB and Competitor sell equal volumes
- Insights
 - Big takeaway is that Competitor sells more via Doctor's Offices, and BB via Optical Retailers
 - Big Box Discounter = Walmart; Sam's Club
 - Doctor's Offices = your local mom & pop non-chain doctor's office
 - Optical Retailer = Lenscrafters, etc.
 - Interviewee should realize that there may be pricing differences between channels, meaning BB would be less profitable at the same volume due to selling to a less profitable mix of customers. If not, prod by reminding him/her about the problem at hand
 - Better interviewees will proactively hypothesize *why* this could be happening –
 - There are varying degrees of buying power: Walmart purchases in large volume, Doctor's offices are purchasing in small quantities, probably not as business savvy (no procurement department). Lenscrafters is somewhere in the middle.
 - Also, varying pricing to customers: Walmart is known for cheap prices, Doctor's offices are specialized and focus on high service, Lenscrafters is in the middle
 - If interviewee is not forthcoming with why this might be happening, ask along the lines of “does this surprise you?”

BB's Contact Lenses

- Question 3: (show Exhibit 2: Profitability by Distribution Channel) What do you see here? Does any of this surprise you?
 - Interviewee should explain the information shown on the slide, identifying that prices are varying across channels
 - If interviewee moves to fill out the “profit” row, ask them to first explain why revenues and costs vary in this way. Any logical answer is acceptable, e.g.
 - Customers willing to pay a premium at doctor’s office for more individual services
 - COGS would be equal across channels, as it’s the same product
 - Sales, distribution: Possibly advertising spend/ dedicated sales personnel to manage the big box and optical retailer relationships
 - Fixed costs: Same product

BB's Contact Lenses

- Question 3b: Ask interviewee to calculate profit and percent margin by channel. OK to estimate on margin
 - Big Box Discounter: \$3.00 (18%)
 - Doctor's Office: \$16.00 (57%)
 - Optical Retailer: \$8.50 (38%)
- Question 3c: Given this information and given the initial problem you're solving for, what would you want to look at now?
 - Strong interviewees would have moved to the “so what” once they completed the profit and margin calculations and would not need this prompt
 - Good answer: Given the significantly higher margins in the Doctor's Office distribution channel, I would want to look into how we can sell more lenses via the Doctor's Office channel?
 - Better interviewees might also comment that they're curious as to why BB is even bothering to sell in the Big Box Channel – why not pull out and focus resources/attention on improving sales in Doctor's Offices instead?

BB's Contact Lenses

- Question 4:

- Like its Competitor, BB relies on sales reps to distribute its contact lenses to the Doctor's Offices. Currently, BB has 5 reps in its call center dedicated to reaching out to the Doctor's Offices and doing whatever is necessary to get them to sell as many BB lenses as possible. Interestingly, McKinsey has discovered a relationship between call frequency and sales generated:
 - If two calls are made to a doctor's office per month, BB sees a 5% inc in revenue from that office
 - If a 3rd call is made to a doctor's office that month, our client sees an additional 10% increase in revenue from that office
 - There is no additional revenue for calling on a doctor's office beyond this monthly frequency
- Currently, each BB sales person has 100 doctor's office customer accounts. Policy is that each account must be called at least once a month, as BB does not want to lose any customers. Assume that sales per doctor's office, with one sales call per month, is \$100. **Given these findings, what should BB do?**

- Info for Interviewee:

- To maximize sales to doctor's offices, interviewee must consider how best to distribute calls across the 100 customers (see next slide for info/ calculations)
- Clarify if necessary: if any one doctor's office is not called once per month, BB loses the account

BB's Contact Lenses

- Question 4: Math

Information to be provided (if asked):

- Each sales call takes 30 minutes
- Each sales rep works 20 days per month, 8 hours a day
- Each sales rep spends approximately 3 hours of each work day on admin work, breaks, etc
- Each sales rep has 100 accounts
- \$100 base revenue per customer per month
 - 5% increase for 2 calls
 - 10% additional increase for 3 calls

Math:

Available capacity for more calls, per sales rep

- 8 hours/day - 3 hours/day for admin = 5 hours/day for calls
- 5 hours/ 30 min per call = 10 calls per day
- 10 calls/ day * 20 days/ month = Max Capacity: 200 calls
- 200 capacity minus 100 calls necessary (1 per customer) = time for 100 more calls

Each rep already calls each customer once, so must call 2 more times to reach 3/ month.

Interviewee should realize each BB sales rep could either:

- **Call 50 customers 3 times and 50 customers 1 time**
- **Call 100 customers 2 times**

Sales by number of calls

- 1 call = \$100
- 2 calls = \$100 * 1.05 = \$105
- 3 calls = \$105 * 1.10 = \$115.5

Approach to maximize sales:

- If 100 customers are called twice, then sales = 100 customers * \$105 = \$10,500
- Otherwise, sales = 50 customers * \$100 + 50 customers * \$115.5 = \$10,775

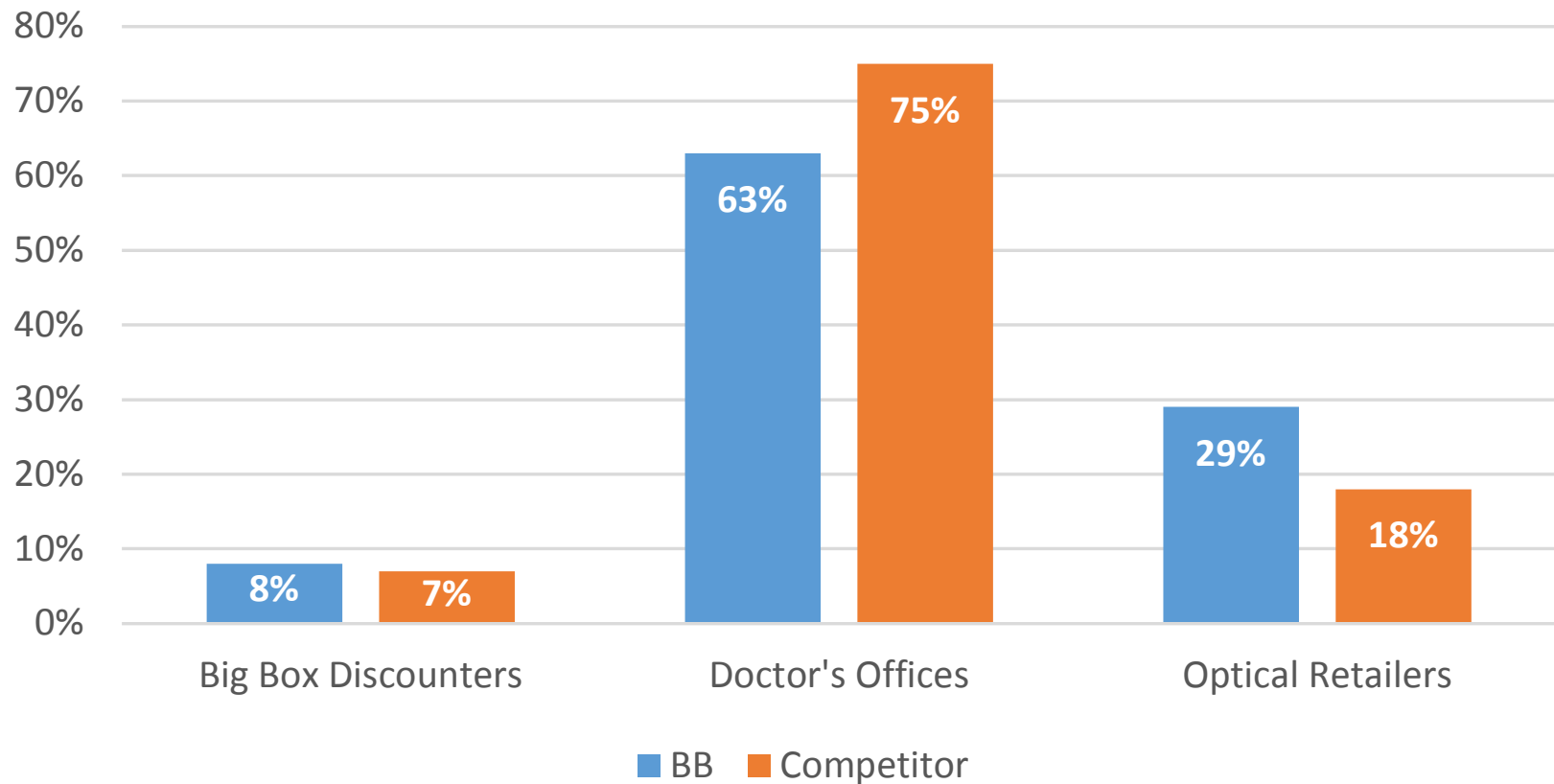
- **To maximize sales, BB should have each sales rep call 50 customers 1x and 50 customers 3x per month. Total sales of \$10,775/ month per sales rep**
- Interviewee could also mention that if the 3 hours of admin work per day could be decreased, sales reps could increase revenue by calling more customers 3 times – with 7.5 hours of call time per day they could make 300 calls per month – or could propose looking into hiring additional sales reps

BB's Contact Lenses

- The CEO of BB walks past you in the hallway and asks what you've come up with so far. What are your recommendations?
 - Range of acceptable answers as long as they are brief and supported by information. An answer could introduce and explain at a high level:
 - Lagging profitability compared to Competitor A likely due to having a stronger presence in less profitable distribution channels; recommendation to increase sales through doctors' offices (maybe through refocusing resources currently driving big-box sales)
 - Additional opportunity to increase revenue from existing doctor's office customers by making 3 sales rep calls per month, as sales per month increases with each additional call up to 3 and policy is currently to make only 1
 - Can increase revenue by reallocating current sales rep resources, and could explore adding more staff to maximize revenue from current customers and acquire new customers in this more profitable distribution channel
 - Exceptional candidates would add risks and next steps, e.g.
 - Potential for the relationship between calls and sales to change over time; should track this ongoing
 - Cost-benefit analysis of adding extra representatives to increase calling frequency to all customers

BB's Contact Lenses: Exhibit 1

Customer Mix



BB's Contact Lenses: Exhibit 2

Profitability by Distribution Channel

Per box of contacts (6 lenses)	Big Box Discounter	Doctor's Office	Optical Retailer
Revenue	\$16.50	\$28.00	\$22.00
COGS	\$8.00	\$8.00	\$8.00
Sales, Distribution	\$3.50	\$2.00	\$3.50
Other Fixed Costs	\$2.00	\$2.00	\$2.00
Profit			

Case 7:

Emerging Markets



Firm Style	Interview Round
BCG	Final

Case Question

The government of an emerging market country has recently approached our client, PharmaCo, a large pharmaceuticals company, to apply to manufacture a vaccine for their citizens. To win the bid, the government requires that the vaccine be manufactured in their country. The government will not decide which company gets the contract until after all bidders have an in-country plant. Should our client make the investment to build a manufacturing plant?

Clarifying information *(provide only if corresponding questions are asked)*

What is the primary objective for the client?	Profit, though other considerations may be important
What other products does our client have? / Is this our client's only product?	PharmaCo's current product is an older product with declining sales in developed markets
Who are our client's competitors for this bid?	PharmaCo has two primary competitors: • Competitor A's product is a new innovation. Costs more to produce and is currently lower quality than PharmaCo's product • Competitor B's product is similar in product lifecycle, price, manufacturing cost, etc to PharmaCo's product. Competitor B also has comparable size, scale and brand recognition

Emerging Markets

- Framework/ Structure – a good framework could include:
 - Investment decision
 - Revenues
 - Number eligible for vaccine, market penetration, any potential substitutes, growth of customer base
 - Price to customers
 - Costs
 - Vaccine production – raw materials, labor
 - Plant investment cost
 - Likelihood of winning bid
 - Selection criteria
 - Competitors
 - Likelihood to enter bid
 - Performance against criteria
 - PharmaCo performance against criteria
 - PharmaCo Capabilities
 - International operating experience?
 - Financial situation – able to take on risk of project and up-front investment?
 - Company brand

Emerging Markets

- Question 1: What would you advise the client in an absence of competition (certainty of winning bid)
 - Interviewee should recognize that this is an NPV analysis – prompt if not

Information to be provided (if asked):

One-time plant investment: \$50M

Country population: 60M, zero population growth

- Assume life expectancy of 60 years and the population is evenly distributed across age groups
- Vaccine is given one time to children at age 2; requires 3 doses for immunity

Guaranteed price: \$4/dose

All-in cost to produce: \$1/dose

Assuming the company gets the contract, they are guaranteed a lifetime patent

The company uses a 10% discount rate in perpetuity

Math:

60M total population across 60 yr life expectancy = 1M children age 2 receiving vaccines each year

Revenue: 1M vaccines * 3 doses/ vaccine * \$4/dose = \$12M revenue per year

Cost: 1M vaccines * 3 doses/ vaccine * \$1/dose = \$3M cost per year

Profit: \$12M - \$3M = \$9M per year

Perpetuity value = \$9/10% = \$90M

NPV = \$90M - \$50M investment = \$40M

If the client is guaranteed to receive the contract, positive NPV of \$40M

Emerging Markets

- Question 2: Given competitive landscape, is the investment still attractive?
 - Share that the government will make a decision based on 1) ability to meet the \$4/dose price guidance and 2) quality
 - Interviewee should recognize that competitor A's higher-cost, lower-quality product is not competitive in the bidding process
 - Interviewee should focus on competitor B – provide the information that competitor B has announced the intention to bid and, because they have similar quality and economics to PharmaCo, they have a 50% chance of winning
 - Interviewee should realize that PharmaCo's NPV is now negative, given the \$50M required investment and 50% chance of winning the bid
 - NPV: $\$90\text{M value} \times 50\% = \$45\text{M expected value} - \$50\text{M investment cost} = -\5 NPV
 - This can be set up as a prisoner's dilemma (next slide)
- Insights
 - If PharmaCo can manage their costs, they can make the investment decision more attractive
 - interviewee should recognize and explain this; interviewer can then share that it is not possible for the client to achieve a better NPV situation through managing costs
 - Or, if there is a way to see whether Competitor B is going to make a bid before committing, Competitor A

Emerging Markets

- Question 2 (optional): Interviewee could communicate NPV for each player in the different bid scenarios using a 2x2 matrix:

		PharmaCo	
		Bid	No bid
Competitor B	Bid	PharmaCo: -\$5M Comp B: -\$5M	PharmaCo: \$0M Comp B: \$40M
	No bid	PharmaCo: \$40M Comp B: \$0M	PharmaCo: \$0M Comp B: \$0M

Emerging Markets

- Question 3: What recommendation should we give PharmaCo?
 - Interviewee can provide any recommendation as long as it is backed up with data and logic
 - Sample recommendation: Client should not enter market
 - NPV of -\$5M with a 50% chance of winning bid, NPV of profits \$90M and investment of \$50M
 - Risks:
 - Not entering a bid allows competitor to lock in the full \$40M NPV
 - If competitor B also chooses not to bid, profit is left on the table
 - Lose opportunity to enter new market, potentially expand to other products – this value is not included in NPV analysis
 - Next steps:
 - Explore risks further, particularly strategic importance of this market
 - This vaccine seems to have value; consider looking for similar emerging markets while Competitor B is working on this bid

Case 8:

Cars, Cars, Cars



Firm Style	Interview Round
A.T. Kearney	1

Case Question

Your client is a major car manufacturer. They have hired us to examine their operating model and identify ways to drive an increase in profitability. What kinds of things would you look at?

Clarifying information *(provide only if corresponding questions are asked)*

This case prompt is intentionally open and ambiguous. Multiple clarifying questions are not encouraged – interviewee should go quickly to developing a framework.

- | | |
|---|---|
| • Where do they operate? | • Sales globally; manufacturing located in the US |
| • Do they have a profitability target? | • No, just looking to increase profitability |
| • What is their primary customer segment? | • They have offerings across all consumer groups |
| • Any other questions | • We can discuss that later in the case |

Cars, Cars, Cars

- Framework/ Structure

- WEAK: Mentions revenue and cost, with some general thoughts on how profits are generated
- GOOD: Clearly breaks out revenues and costs and uses industry-specific language, e.g.
 - Revenues
 - Price: Price charged to dealerships (not to consumers)
 - Volume: Number sold in a given year
 - Mix: Models and makes of cars
 - Costs
 - Variable: Raw materials, labor, shipping to dealerships, etc
 - Fixed: Manufacturing overhead, corporate offices, SG&A, etc
- GREAT: Any logical framework that comprehensively covers the manufacturing value chain and additionally describes manufacturing overhead as a major component of costs
 - Could include materials purchase, production costs, storage prior to delivery

Cars, Cars, Cars

- Question 1: The client has decided that we should focus on one particular plant in the Southeast. This is the client's only plant producing cars for sale in international markets, and it has the lowest profitability of any of the client's US plants. What should we look at for this plant specifically to find opportunities to increase its profitability?
 - Interviewer: **after the interviewee identifies each area below**, provide the given information.
 1. Location of plant: Plant is located next to a major seaport
 2. Layout of plant: Plant is production facility plus major storage lots for produced cars
 3. Manufacturing costs: In line with other plants. No opportunity for cost-savings
 4. Product mix/ opportunities to combine or centralize production across plants: Not possible, as the models sold internationally are different from those sold domestically. Cannot sell these models in US
 5. Shipping: Cars must be shipped to destination markets – they are loaded onto ships at the nearby seaport
 6. Storage: Prior to shipment, cars must be stored until they can be loaded onto the ship
 7. Pricing: Prices are in line with competition in each market. The client doesn't see any opportunity to raise prices.
 - Note to interviewer: Interviewee should focus on what makes this plant different from others. The key for the interviewee to realize is that this plant ships to international markets, rather than domestic, and that is resulting in storage costs prior to shipping. For anything not on this list, give an answer that explains why it's not relevant

Cars, Cars, Cars

- Question 1 Insights:

- Stronger interviewees will generate insights without prompting
- Key to realize is that, since manufacturing costs are the same as other plants and there is no opportunity to increase prices, focus should be on storage and shipping costs (biggest area where this plant differs from the other US plants). Interviewee should suggest next steps to investigate these areas
- Once storage and shipping costs have been identified, interviewee should reflect on key drivers of those costs – e.g. storage costs may be dependent on volume and amount of time spent in storage; if there is a permanent storage space it is likely a fixed cost regardless of volume up to a certain capacity

Cars, Cars, Cars

- Question 2: We realized after visiting the plant that they have large numbers of cars sitting in the plant's storage lot waiting to be loaded onto ships. What should we look for here?
 - Interviewer – provide the numerical information below *after the interviewer asks for it*. Do not prompt calculations, but if necessary can prompt deeper analysis with questions like “Should we look at this on a per-foot basis?”

Information to be provided (if asked):

- **Total capacity of storage lot:** 1,000,000 sq ft
- **Overhead costs:** \$.20 per sq ft per week
- **Car storage usage:** Each car takes up 100 sq ft
- **Length of storage:** Each car remains in storage for an average of 1 week
- **Production volume:** 5000 cars per week
- **Overcapacity charges:** If necessary, extra space costs \$.25 per sq ft per week

Math:

- **Lot capacity:** $1\text{M sq ft} / 100 \text{ sq ft per car} = 10,000 \text{ cars}$
 - Interviewee can then dismiss overcapacity charges at this production volume
- **Overhead costs per week:** $1\text{M sq ft} * \$.20 \text{ per sq ft} = \200K per week
- **Overhead costs per year:** $\$200\text{K} * 52 \text{ weeks} = \10.4M/yr
- **Allocated cost per car :** $\$200\text{K overhead per week} / 5000 \text{ cars produced per week} = \40

• Insights:

- Interviewee should recognize that **regardless of # of cars in storage, the total cost is the same. The overhead cost is expressed in \$/ft but is actually a fixed cost.** Therefore, the higher the production volume, the lower the cost per car.
- Currently, half the storage capacity is empty, resulting in overhead costs to maintain unused storage space. The overcapacity constraint would come into play at 10,000 cars and would impose additional costs, but overall storage cost per car would decrease if the plant increased production volume
- Interviewee may suggest looking into ways to monetize the extra space – this is a great thought, but once they have talked through their logic, tell them the client is not interested in doing this

Cars, Cars, Cars

- Question 3: The client has notified us that they were thinking of altering production levels at the plant, and asked us to advise which option would be most cost-effective. Which would you recommend?
 - Interviewer: Provide Exhibit 1, and then allow interviewee to talk through recommendation using qualitative and quantitative considerations. Math is on next slide
 - Clarifying information to provide if asked:
 - Client believes that increased production would enable partnership with more international dealerships in their current market and would not drive down prices – e.g. additional production would be sold at the client’s current prices – and further investigation into markets, competitors, pricing, etc, is out of scope of this project
 - Average time in storage increases due to necessary shift in shipping timing; however, would expect economies of scale in the shipping cost from increased volume, though do not have the expected shipping costs at this stage
 - Interviewee should be able to explain Exhibit 1 and see immediately that if the cars are in storage for more than a week on average, there is a concern about hitting the capacity constraint

Cars, Cars, Cars

- Question 3 Math:

	Option 1	Option 2	Option 3
Average # cars in storage at any point	3,600 cars per week, in storage for 4 days (~.5 week) = ~1,800 cars	5,000 cars per week, in storage for 7 days = 5,000 cars	8,200 cars per week, in storage for 10 days = $8,200 * 10/7 = \sim 12,000$ cars: OVER CAPACITY
Total storage cost per week	\$200,000 No overcapacity needed	\$200,000 No overcapacity needed	\$200,000 PLUS overcapacity: 2,000 extra cars * 100 sq ft = 200,000 sq ft overcapacity needed 200,000 sq ft * \$.25/sq ft = \$50,000 \$200,000+\$50,000 = \$250,000
Allocated storage cost per car	\$200,000 per week / 3,600 cars per week = \$55.56	\$200,000 / 5,000 cars = \$40	\$250,000 / 8,200 cars = \$30.49

Cars, Cars, Cars

- Question 3 Insights:

- A good interviewee will not need to be reminded that the client's objective was **cost-effectiveness**
- Interviewee should identify that options 1 and 2 will have the same total storage cost, but the cost allocated per car varies
 - Option 2 keeps production and costs the same as the current state
 - Option 1 will clearly have the highest cost per car as it is the same total cost with fewer cars per week – interviewee does not need to do this math if this explanation is provided
- Option 3 has a higher total cost of storage, but the lowest cost allocated per car
 - In terms of the client's goal of cost-efficiency, assuming that other costs are held constant and the client can sell the additional volume at current prices, Option 3 is the most attractive
- The decrease in per-car allocated storage cost and the TBD decrease in shipping cost would increase margins for this plant

Cars, Cars, Cars

- Question 4: The client is on the phone asking what production decision they should make. What will you tell them?
 - Note to interviewer: DON'T allow interviewee a moment to prepare an answer
 - A strong interviewee will provide the recommendation in the first sentence, support with the cost-savings information and acknowledge risks.
 - Option 3 is most compelling given the cost-savings information
 - Risks and next steps for Option 3 could include:
 - Selling the additional production volume: how confident is the client in this and will they be able to maintain current prices? Further investigation needed here
 - Increasing production at the factory – given no information about production capacity or costs, it is unclear whether there would be cost savings or additional costs to accommodate this, and next steps are to investigate
 - Fluctuating sales could mean that inventory levels might be too low or high to meet demand; delays in shipping due to fluctuating sales would incur additional overcapacity storage charges

Cars, Cars, Cars: Exhibit 1

- Plant production and shipping options identified by client:

	Production Volume per week	Average Time in Storage
Option 1	3,600	4 days
Option 2	5,000	7 days
Option 3	8,200	10 days

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